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Ontario Heart Monitoring Analytics

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BIG DATA 2 ASSIGNMENT

Content Outline

Topics for discussion

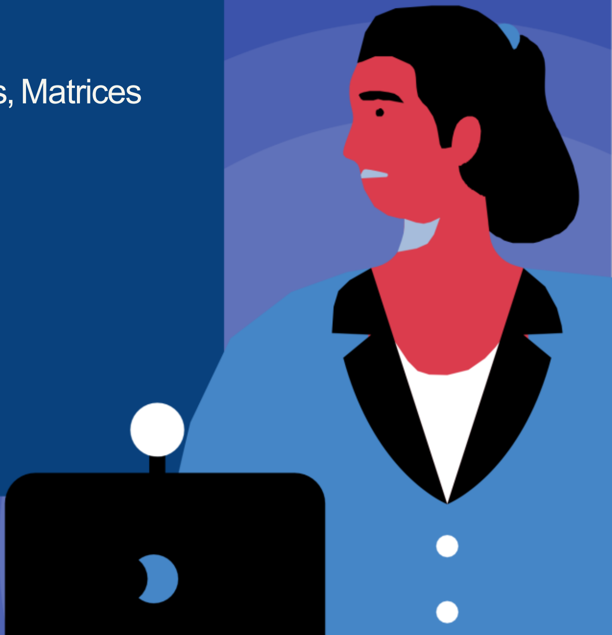
01 Introduction to the Company

02 Data Set from 2010 to 2016

03 Data Creation Using : Vectors, Matrices and Data Frames

04 Analysis: DPLYR Library

05 Visualization: Graphical Representation



OHMA

Data Saves Lives.

Ontario Heart Monitoring Analytics (OHMA) was founded in 2015 by 5 classmates working on a project. Little did they know that 6 years down the lane , OHMA would play a vital part in helping the people of Ontario to lead a healthier lifestyle.

OHMA runs on the basis of gathering heart diseases related data by collecting information depending on gender, age , year , population.

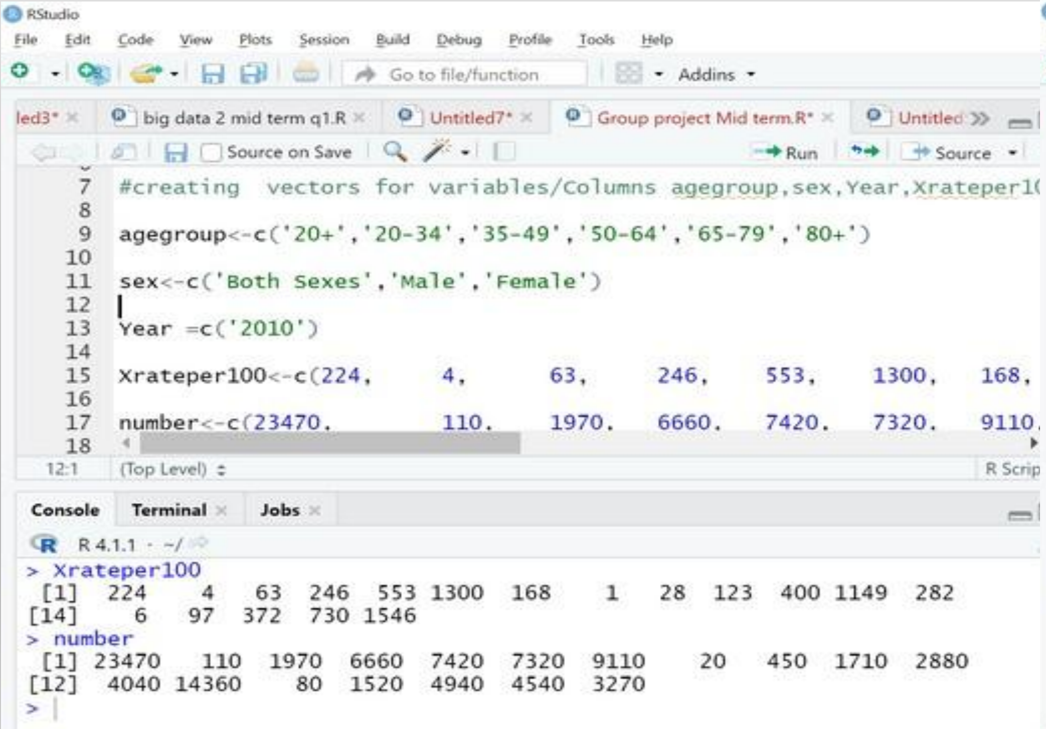
Dataset (2010 – 2016)

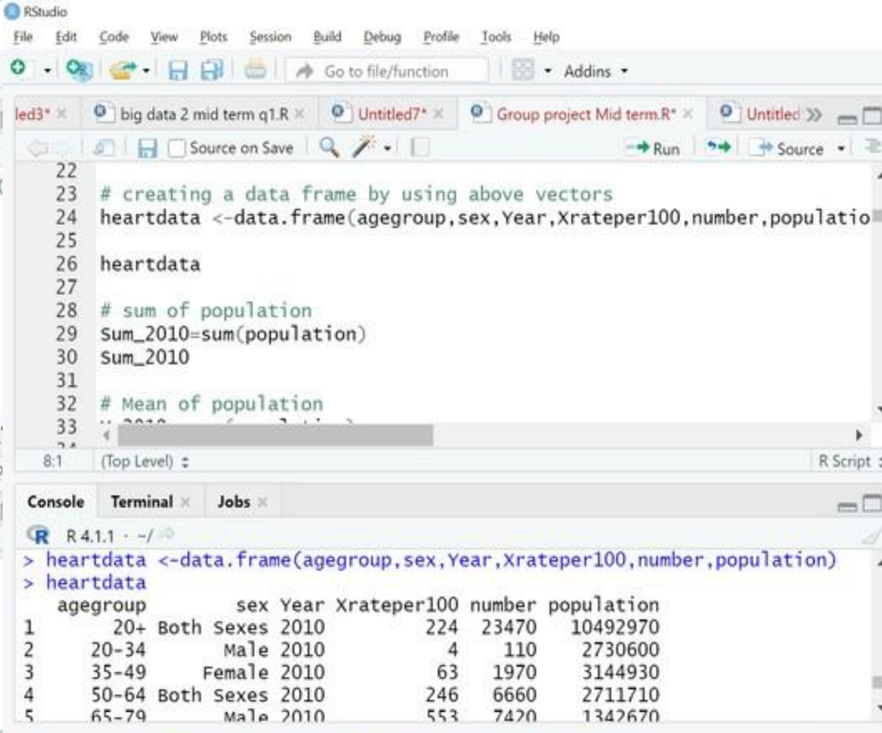
	A	B	C	D	E	F	G	H
1	Age Group	Sex	Year	Rate (per 100,000)	Number	Population		
2	'20+'	Both sexes	2010	224	23,470	10,492,970		
3	'20-34'	Both sexes	2010	4	110	2,730,600		
4	'35-49'	Both sexes	2010	63	1,970	3,144,930		
5	'50-64'	Both sexes	2010	246	6,660	2,711,710		
6	'65-79'	Both sexes	2010	553	7,420	1,342,670		
7	'80+'	Both sexes	2010	1,300	7,320	563,080		
8	'20+'	Females	2010	168	9,110	5,408,000		
9	'20-34'	Females	2010	1	20	1,369,270		
10	'35-49'	Females	2010	28	450	1,581,130		
11	'50-64'	Females	2010	123	1,710	1,385,210		
12	'65-79'	Females	2010	400	2,880	720,770		
13	'80+'	Females	2010	1,149	4,040	351,620		
14	'20+'	Males	2010	282	14,360	5,084,970		
15	'20-34'	Males	2010	6	80	1,361,320		
16	'35-49'	Males	2010	97	1,520	1,563,800		
17	'50-64'	Males	2010	372	4,940	1,326,480		
18	'65-79'	Males	2010	730	4,540	621,900		
19	'80+'	Males	2010	1,546	3,270	211,460		
20	'20+'	Both sexes	2011	213	22,700	10,674,590		
21	'20-34'	Both sexes	2011	4	110	2,779,460		

heartdata

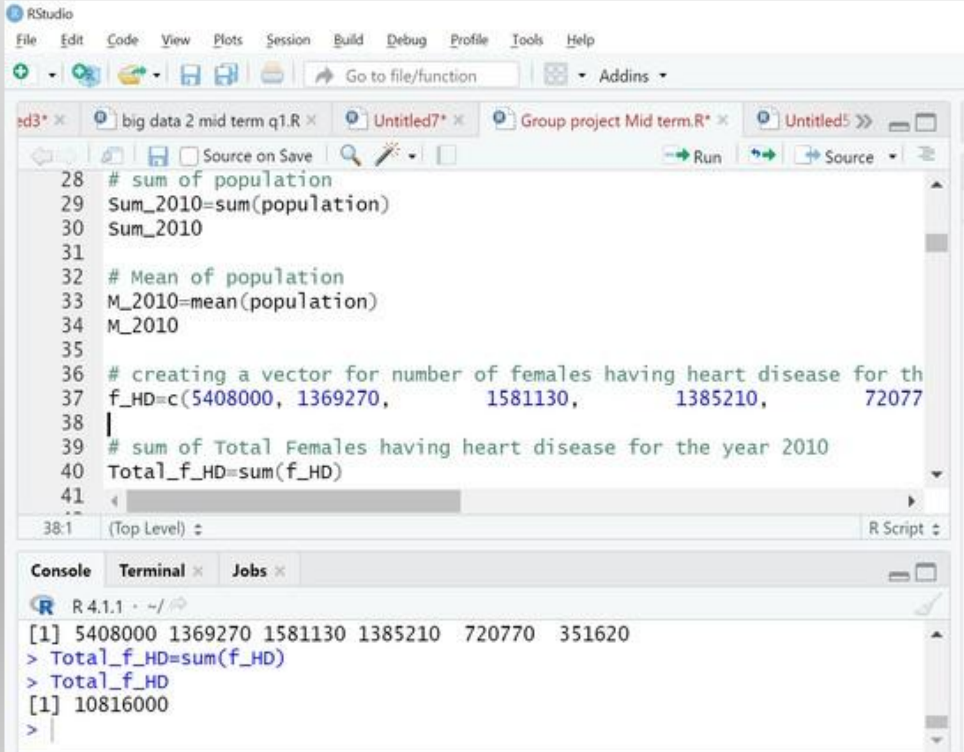
Ready

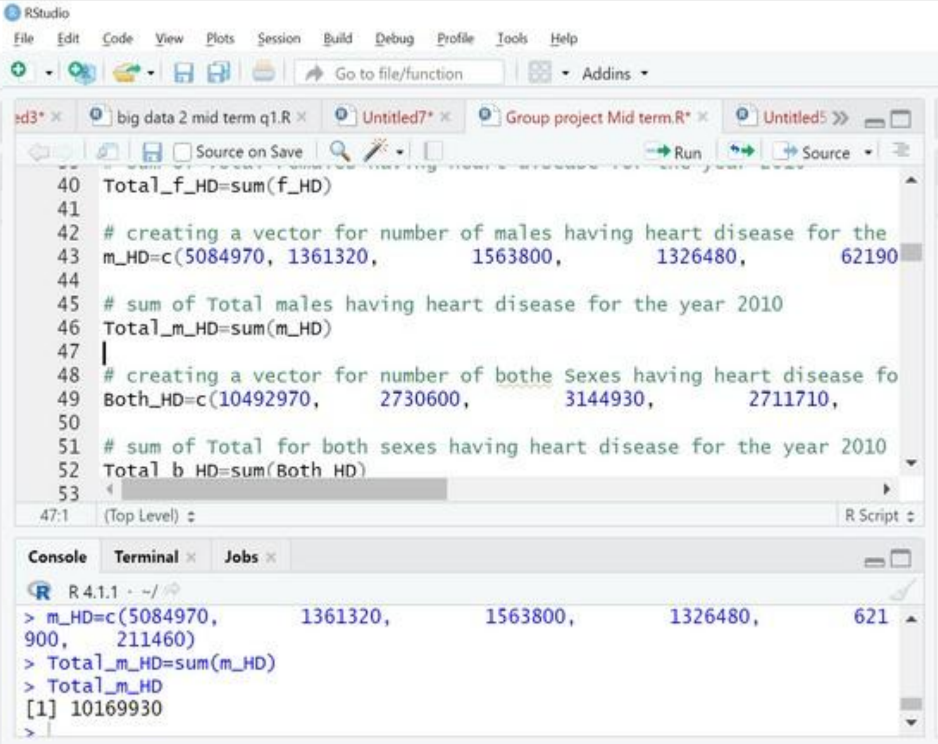
Creating Vectors and the Data Frame



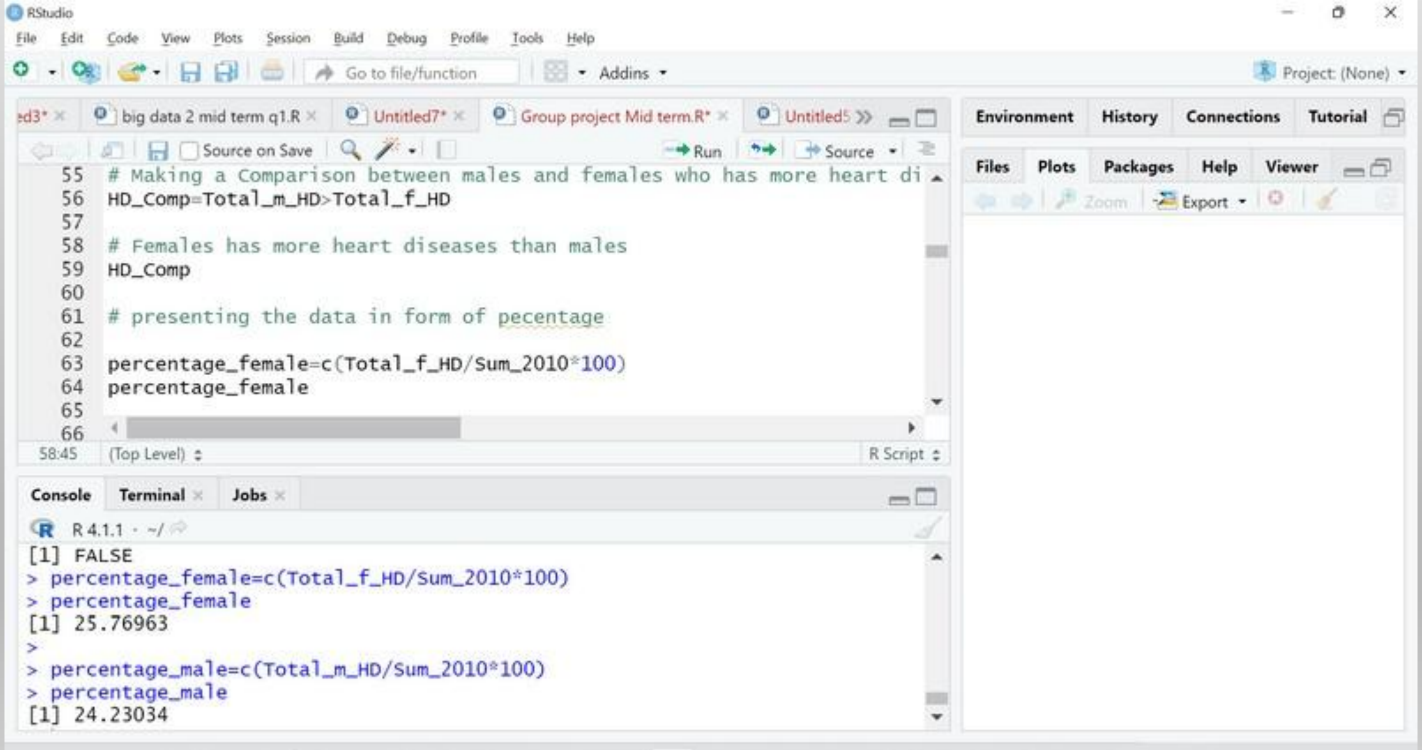


Using Sum function on Vector for Females and Males

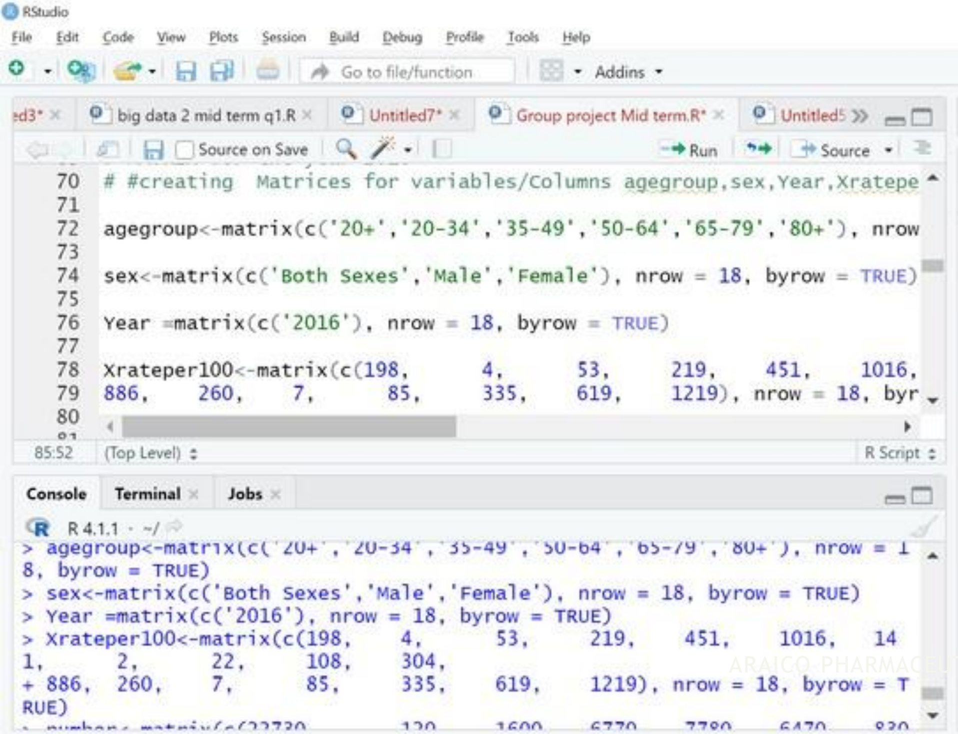


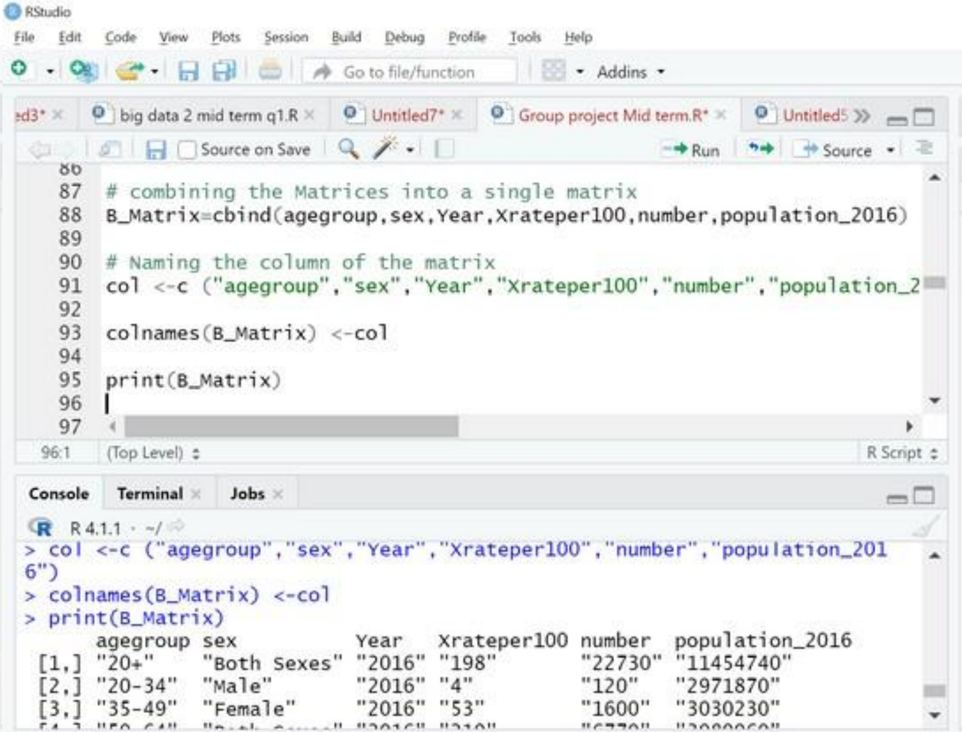


Making Comparison between males and females



Creating a new single Matrix by Combining Column Matrices





DPLYR

By Using Select Function output

```r
mydata2
# Showing data without  Year and Population columns
mydataminus = select(mydata, -Year, -Population)
mydataminus
# Selecting Variables contain 'X' in their columns names
```

12:40 (Top Level)

Console Terminal Jobs

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Smester 3/big data 2/

```
126 2016       249310
> mydataminus = select(mydata, -Year, -Population)
> mydataminus
   Age.Group        Sex X.Rate..per.100.000.. Number
1      '20+' Both sexes                   224  23470
2    '20-34' Both sexes                     4    110
3    '35-49' Both sexes                    63   1970
4    '50-64' Both sexes                   246   6660
5    '65-79' Both sexes                   553   7420
6      '80+' Both sexes                  1300   7320
7      '20+'    Females                   168   9110
8    '20-34'    Females                     1     20
9    '35-49'    Females                    28    450
10   '50-64'    Females                   123   1710
11   '65-79'    Females                   400   2880
12     '80+'    Females                  1149   4040
13     '20+'      Males                   282  14360
14   '20-34'      Males                     6     80
```

```r
# Selecting Variables contain 'B' and star with 'X' in their columns name
mydata_X = select(mydata, starts_with("X"))
mydata_X
mydata_B = select(mydata, contains("B"))
mydata_B
```

12:9 (Top Level)

Console Terminal Jobs

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Smester 3/big data 2/

```
   X.Rate..per.100.000..
1                    224
2                      4
3                     63
4                    246
5                    553
6                   1300
7                    168
8                      1
9                     28
10                   123
11                   400
12                  1149
13                   282
14                     6
15                    97
16                   372
17                   730
18                  1546
19                   213
20                     4
21                    59
22                   223
23                   519
24                  1243
```

Using Arrange and Distinct function

Group project Mid term.R* | Untitled13* | pie chart.R | heart_data_ontario11

Source on Save

```r
mydata_B

# using Arrange function
arranage_mydata=arrange(mydata,desc(Year))
```

17:1 (Top Level)

Console Terminal Jobs

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Smester 3/big data 2/

```
> arranage_mydata
   Age.Group         Sex Year X.Rate..per.100.000.. Number Population
1      '20+' Both sexes 2016                   198  22730   11454740
2     '20-34' Both sexes 2016                     4    120    2971870
3     '35-49' Both sexes 2016                    53   1600    3030230
4     '50-64' Both sexes 2016                   219   6770    3089960
5     '65-79' Both sexes 2016                   451   7780    1726160
6      '80+' Both sexes 2016                  1016   6470     636520
7      '20+'    Females 2016                   141   8300    5897690
8     '20-34'    Females 2016                     2     30    1474170
9     '35-49'    Females 2016                    22    340    1540860
10    '50-64'    Females 2016                   108   1690    1571070
11    '65-79'    Females 2016                   304   2810     924360
12     '80+'    Females 2016                   886   3430     387210
13     '20+'      Males 2016                   260  14430    5557050
14    '20-34'      Males 2016                     7    100    1497710
15    '35-49'      Males 2016                    85   1260    1489370
16    '50-64'      Males 2016                   335   5090    1518890
17    '65-79'      Males 2016                   619   4960     801790
```

Source on Save

```r
#Below we are using two variables Sex and Year to determine uniqueness
x5 = distinct(mydata, Sex, Year, keep_all= TRUE)
```

20:7 (Top Level)

Console Terminal Jobs

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Smester 3/big data 2/

```
4 2013      TRUE
5 2014      TRUE
6 2015      TRUE
7 2016      TRUE
> x5 = distinct(mydata, Sex, Year, keep_all= TRUE)
> x5
          Sex Year keep_all
1  Both sexes 2010     TRUE
2     Females 2010     TRUE
3       Males 2010     TRUE
4  Both sexes 2011     TRUE
5     Females 2011     TRUE
6       Males 2011     TRUE
7  Both sexes 2012     TRUE
8     Females 2012     TRUE
9       Males 2012     TRUE
10 Both sexes 2013     TRUE
11    Females 2013     TRUE
12      Males 2013     TRUE
13 Both sexes 2014     TRUE
```

Filter, Pipe and Grepl function

```r
mydata7 = filter(mydata, Sex == "Males")
# using Pipe and filter getting data of specific column
filter_spec = filter(mydata, Sex %in% c("Males") & Year %in%c("2010") & Age.Group %in% c("'80+
filter_spec
# using Filter and greple function to show value in population column having figure 23
mydata10 = filter(mydata,grepl("23",Population))
```

41:1 (Top Level) R Script

Console Terminal Jobs

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Smester 3/big data 2/

```
32    '20-34' Males 2015                      5    80    1471510
33    '35-49' Males 2015                     88  1300    1484250
34    '50-64' Males 2015                    328  4910    1497500
35    '65-79' Males 2015                    624  4820     772470
36      '80+' Males 2015                   1239  2970     239670
37      '20+' Males 2016                    260 14430    5557050
38    '20-34' Males 2016                      7   100    1497710
39    '35-49' Males 2016                     85  1260    1489370
40    '50-64' Males 2016                    335  5090    1518890
41    '65-79' Males 2016                    619  4960     801790
42      '80+' Males 2016                   1219  3040     249310
> mydata8 = filter(mydata, Sex %in% c("Males") & Year %in%c("2010") & Age.Group %in% c("'80+'"))
> filter_spec = filter(mydata, Sex %in% c("Males") & Year %in%c("2010") & Age.Group %in% c("'80
+'"))
> filter_spec
  Age.Group   Sex Year X.Rate..per.100.000.. Number Population
1     '80+' Males 2010                  1546   3270     211460
> mydata10 = filter(mydata,grepl("23",Population))
> mydata10
  Age.Group       Sex Year X.Rate..per.100.000.. Number Population
1   '20-34'   Females 2012                     1     20    1411230
2     '80+'     Males 2013                  1333   3160     237120
3     '80+'     Males 2014                  1350   3180     235490
```

Adding New Variable By Using Mutate Function

```r
# percentage of affcted people from heart disease by adding new variable
percent_Ag=mydata %>% select(Number,Population) %>% mutate(mydata,percent_Ag=(Number/Populatio
percent_Ag
```

61:11 (Top Level) R Script

Console Terminal Jobs

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Smester 3/big data 2/

```
> percent_Ag=mydata %>% select(Number,Population) %>% mutate(mydata,percent_Ag=(Number/Population*1
00))
> percent_Ag
   Number Population Age.Group         Sex Year X.Rate..per.100.000..  percent_Ag
1   23470   10492970     '20+'  Both sexes 2010                   224 0.223673564
2     110    2730600   '20-34'  Both sexes 2010                     4 0.004028419
3    1970    3144930   '35-49'  Both sexes 2010                    63 0.062640504
4    6660    2711710   '50-64'  Both sexes 2010                   246 0.245601484
5    7420    1342670   '65-79'  Both sexes 2010                   553 0.552630207
6    7320     563080     '80+'  Both sexes 2010                  1300 1.299992896
7    9110    5408000     '20+'     Females 2010                   168 0.168454142
8      20    1369270   '20-34'     Females 2010                     1 0.001460632
9     450    1581130   '35-49'     Females 2010                    28 0.028460658
10   1710    1385210   '50-64'     Females 2010                   123 0.123446986
```

Using Rank Function In Descending Order

```r
mydata$Number_rank=rank(mydata$Number)
mydata$Number_rank

# Using Rank Function with Descending  order on Year Column
mydata$Year_rank=rank(desc(mydata$Year))
mydata$Year_rank
```

```
177:1   (Top Level) ÷                                                    R Script

Console   Terminal ×   Jobs ×

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Smester 3/big data 2/
123    1260     1489370    '35-49'       Males  2016                     85  0.084599529
124    5090     1518890    '50-64'       Males  2016                    335  0.335113142
125    4960      801790    '65-79'       Males  2016                    619  0.618615847
126    3040      249310     '80+'        Males  2016                   1219  1.219365449
> mydata$Number_rank=rank(mydata$Number)
> mydata$Number_rank
  [1] 126.0  16.5  49.0  92.0 100.0  99.0 112.0   2.5  28.0  44.0  53.0  70.0 117.5   9.0  35.0
 [16]  82.0  72.0  63.0 121.0  16.5  48.0  85.0  98.0  97.0 109.5   6.0  26.5  36.0  50.0  69.0
 [31] 113.0   9.0  34.0  74.0  71.0  61.0 123.0  16.5  47.0  86.0 101.0  96.0 111.0   2.5  26.5
 [46]  38.5  56.0  68.0 114.0  11.0  33.0  75.0  73.0  62.0 125.0  20.0  46.0  90.0 103.0  95.0
 [61] 109.5   2.5  25.0  42.0  55.0  67.0 116.0  13.0  32.0  79.0  76.0  59.0 124.0  20.0  42.0
 [76]  89.0 105.0  94.0 108.0   2.5  23.5  40.0  54.0  66.0 117.5  13.0  31.0  78.0  80.5  60.0
 [91] 120.0  16.5  38.5  91.0 102.0  87.0 107.0   6.0  23.5  45.0  52.0  65.0 115.0   9.0  30.0
[106]  80.5  77.0  57.0 122.0  20.0  37.0  93.0 104.0  88.0 106.0   6.0  22.0  42.0  51.0  64.0
[121] 119.0  13.0  29.0  84.0  83.0  58.0
> mydata$Year_rank=rank(desc(mydata$Year))
> mydata$Year_rank
  [1] 117.5 117.5 117.5 117.5 117.5 117.5 117.5 117.5 117.5 117.5 117.5 117.5 117.5 117.5 117.5
 [16] 117.5 117.5 117.5  99.5  99.5  99.5  99.5  99.5  99.5  99.5  99.5  99.5  99.5  99.5  99.5
 [31]  99.5  99.5  99.5  99.5  99.5  99.5  81.5  81.5  81.5  81.5  81.5  81.5  81.5  81.5  81.5
 [46]  81.5  81.5  81.5  81.5  81.5  81.5  81.5  81.5  81.5  63.5  63.5  63.5  63.5  63.5  63.5
 [61]  63.5  63.5  63.5  63.5  63.5  63.5  63.5  63.5  63.5  63.5  63.5  63.5  45.5  45.5  45.5
 [76]  45.5  45.5  45.5  45.5  45.5  45.5  45.5  45.5  45.5  45.5  45.5  45.5  45.5  45.5  45.5
 [91]  27.5  27.5  27.5  27.5  27.5  27.5  27.5  27.5  27.5  27.5  27.5  27.5  27.5  27.5  27.5
```

GRAPHS

Pie Chart : Heart Diseases percentage by Sex

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

dpylr tutorial.R | ass mid ali.R | Group project Mid term.R | graphs edward (1).R

Source on Save Run Source

```
pop <- c(Total_f_HD,Total_m_HD,Total_b_HD)
labels <- c("Females", "Males","BothSexes")

piepercent<- round(100*pop/sum(pop), 1)
# Plot the chart.
pie(pop, labels = piepercent, main = "Percentage of HeartDisease By sex",col = rainbow(length(
legend("topright", c("Females", "Males","BothSexes"), cex = 0.8,fill = rainbow(length(pop)))
```

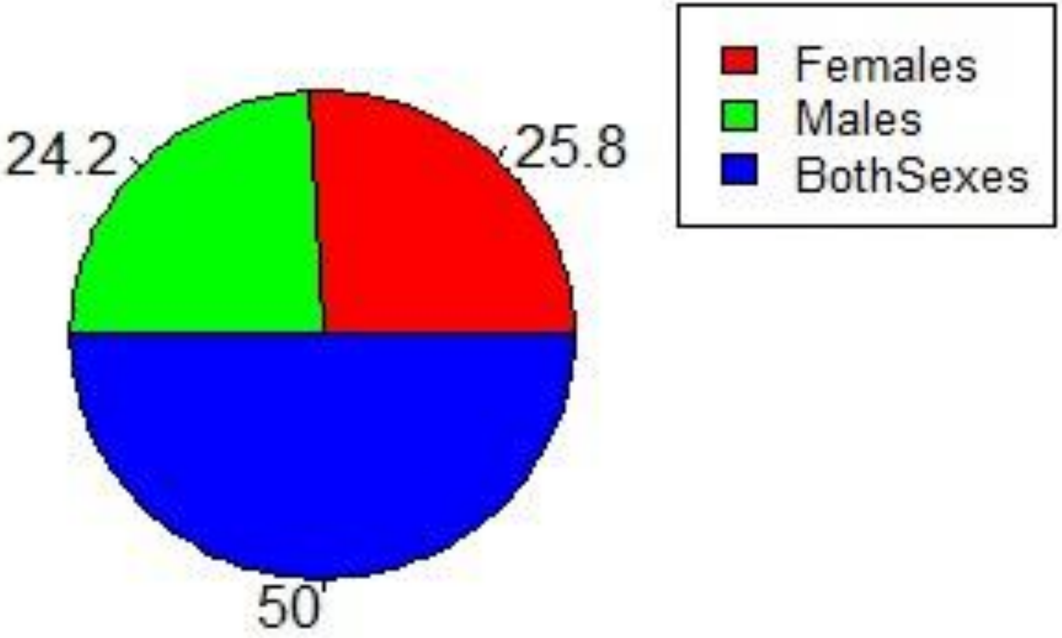
186:1 (Top Level) R Script

Console Terminal Jobs

R 4.1.1 · ~/

```
> pie(pop, labels = piepercent, main = "population pie chart",col = rainbow(length(pop)))
> legend("topright", c("Females", "Males","BothSexes"), cex = 0.8,
+ fill = rainbow(length(pop)))
> # Save the file.
> dev.off()
RStudioGD
        2
>
> mydata = read.csv("C:\\Users\\91873\\Desktop\\Humber\\SEMESTER 3\\Big data 2\\heart _data _ontari
o.csv")
> pop <- c(Total_f_HD,Total_m_HD,Total_b_HD)
> labels <- c("Females", "Males","BothSexes")
> piepercent<- round(100*pop/sum(pop), 1)
> # Plot the chart.
> pie(pop, labels = piepercent, main = "Percentage of HeartDisease By sex",col = rainbow(length(po
p)))
>
```

Perctentage of HeartDisease By sex



Rate of Heart Diseases in Ontario per Year differentiated by Gender

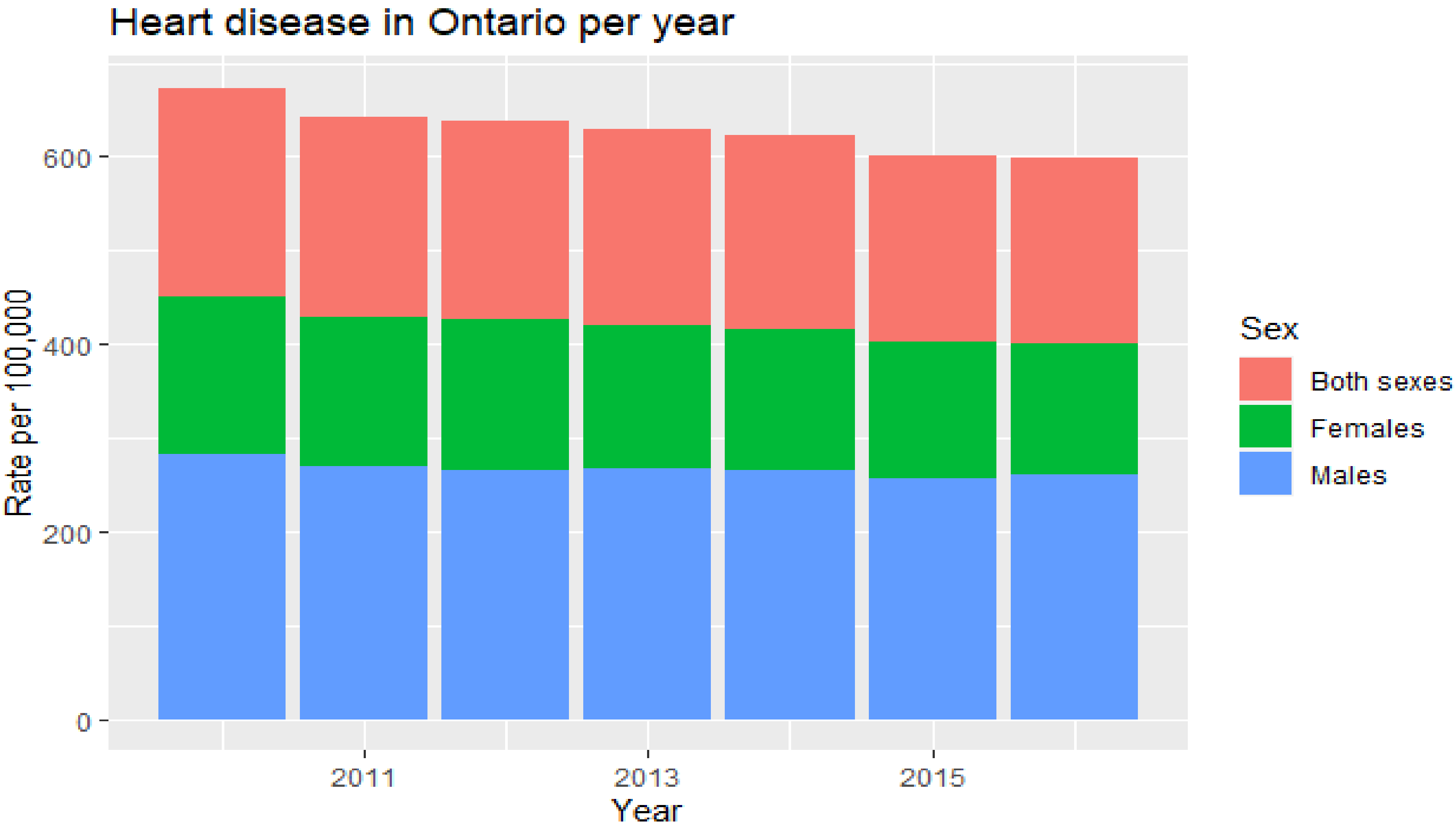
```r
# Ontario Heart Rate data per Year

year <- subset(heartdata, Age.Group == "20+", select = c("Year","Sex","Rate"))

ggplot(year, aes(x=Year, y=Rate, fill=Sex)) +
  geom_histogram(stat='identity')+
  labs(title = "Heart disease in Ontario per year",x="Age Group",y="Rate per 100,000")
```

37:1 (Top Level) R Script

Console Terminal Jobs

R 4.1.1 · ~/

```
+   labs(title = "Heart disease in Ontario per year",x="Age Group",y="Rate per 100,000")
Warning message:
Ignoring unknown parameters: binwidth, bins, pad
> # Ontario Heart Rate data per Year
>
> year <- subset(heartdata, Age.Group == "20+", select = c("Year","Sex", "Rate"))
>
> ggplot(year, aes(x=Year, y=Rate, fill=Sex)) +
+   geom_histogram(stat='identity')+
+   labs(title = "Heart disease in Ontario per year",x="Age Group",y="Rate per 100,000")
Warning message:
Ignoring unknown parameters: binwidth, bins, pad
> total <- subset(heartdata, Age.Group == "20+", select = c("Age.Group","Year","Rate"))
>
> ggplot(total, aes(x=Year, y=Rate)) +
+   geom_bar(stat='identity') +
+   labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
> total <- subset(heartdata, Age.Group == "20+", select = c("Age.Group","Year","Rate"))
>
> ggplot(total, aes(x=Year, y=Rate)) +
+   geom_bar(stat='identity') +
+   labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
> year <- subset(heartdata, Age.Group == "20+", select = c("Year","Sex","Rate"))
>
> ggplot(year, aes(x=Year, y=Rate, fill=Sex)) +
+   geom_histogram(stat='identity')+
+   labs(title = "Heart disease in Ontario per year",x="Age Group",y="Rate per 100,000")
Warning message:
Ignoring unknown parameters: binwidth, bins, pad
>
```



Heart disease in Ontario per year
Rate per 100,000
600
400
200
0
2011
2013
2015
Year
Sex
Both sexes
Females
Males

Total Number of Heart Disease in Ontario (Yearly)

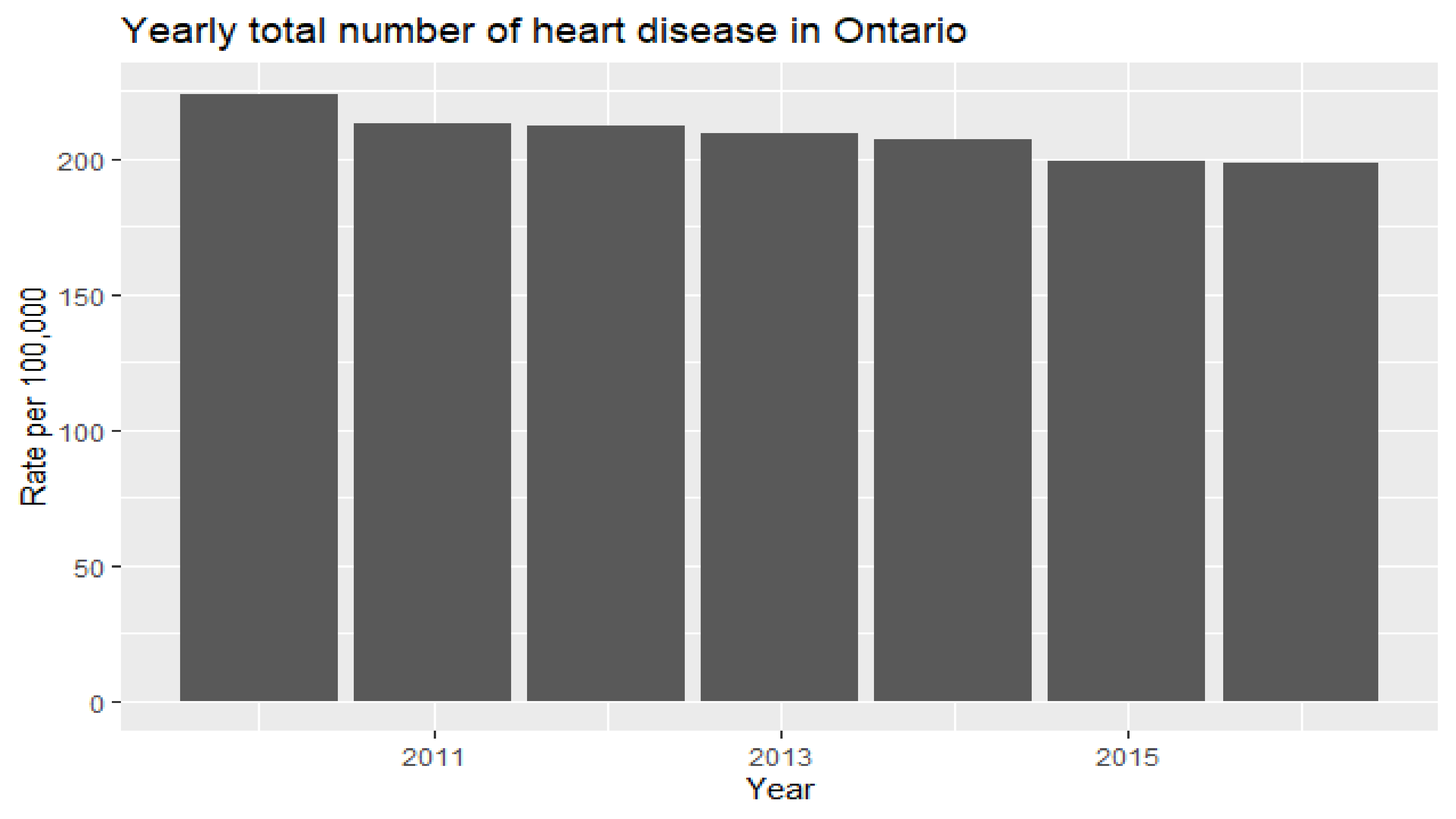
```r
total <- subset(heartdata, Sex=="Both sexes", select = c("Age.Group","Sex","Year","Rate"))

totalYear <- filter(total, Age.Group =="20+" )

ggplot(totalYear, aes(x=Year, y=Rate)) +
  geom_bar(stat='identity') +
  labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
```

40:26 (Top Level) R Script

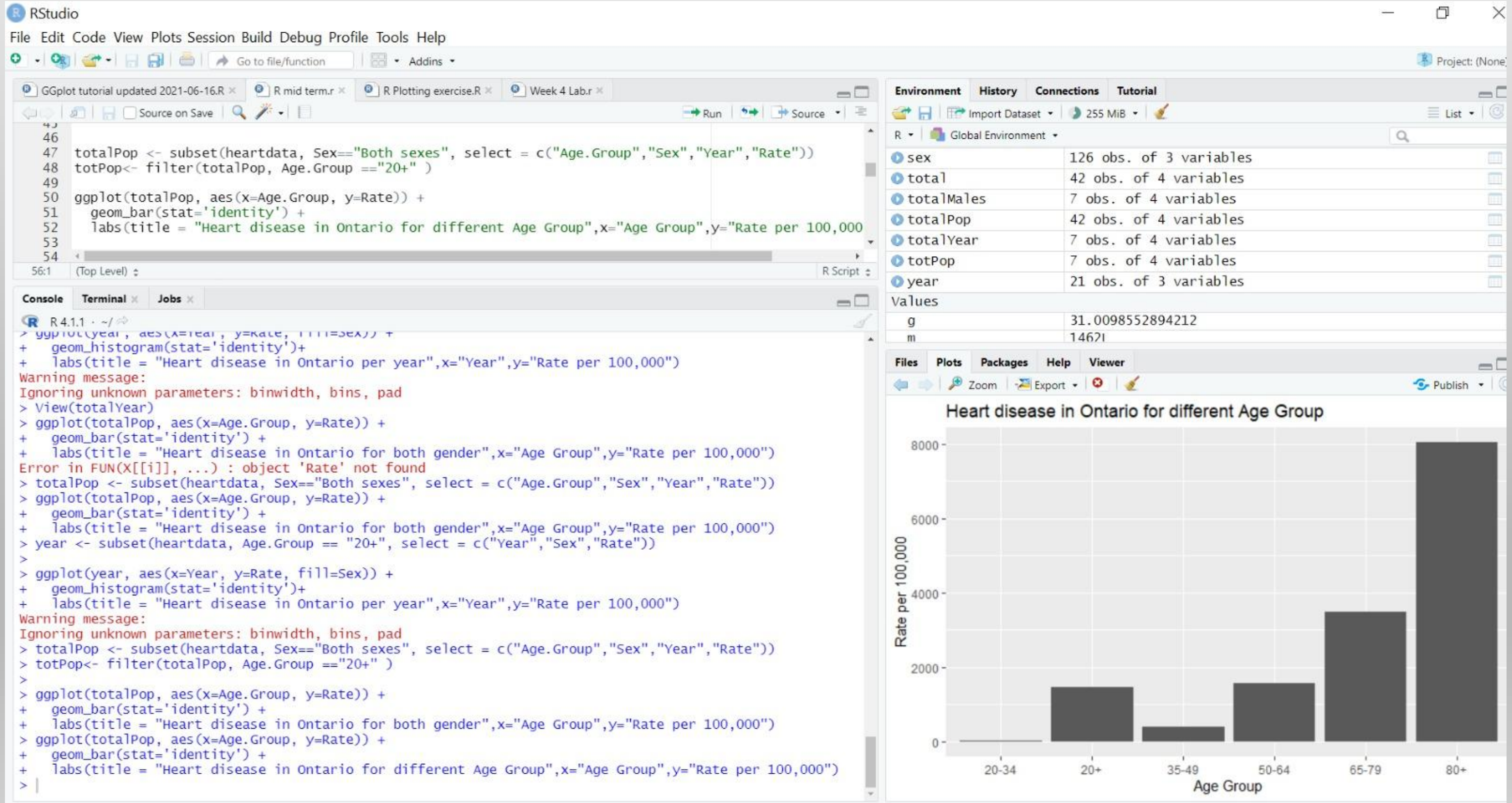
Console Terminal Jobs

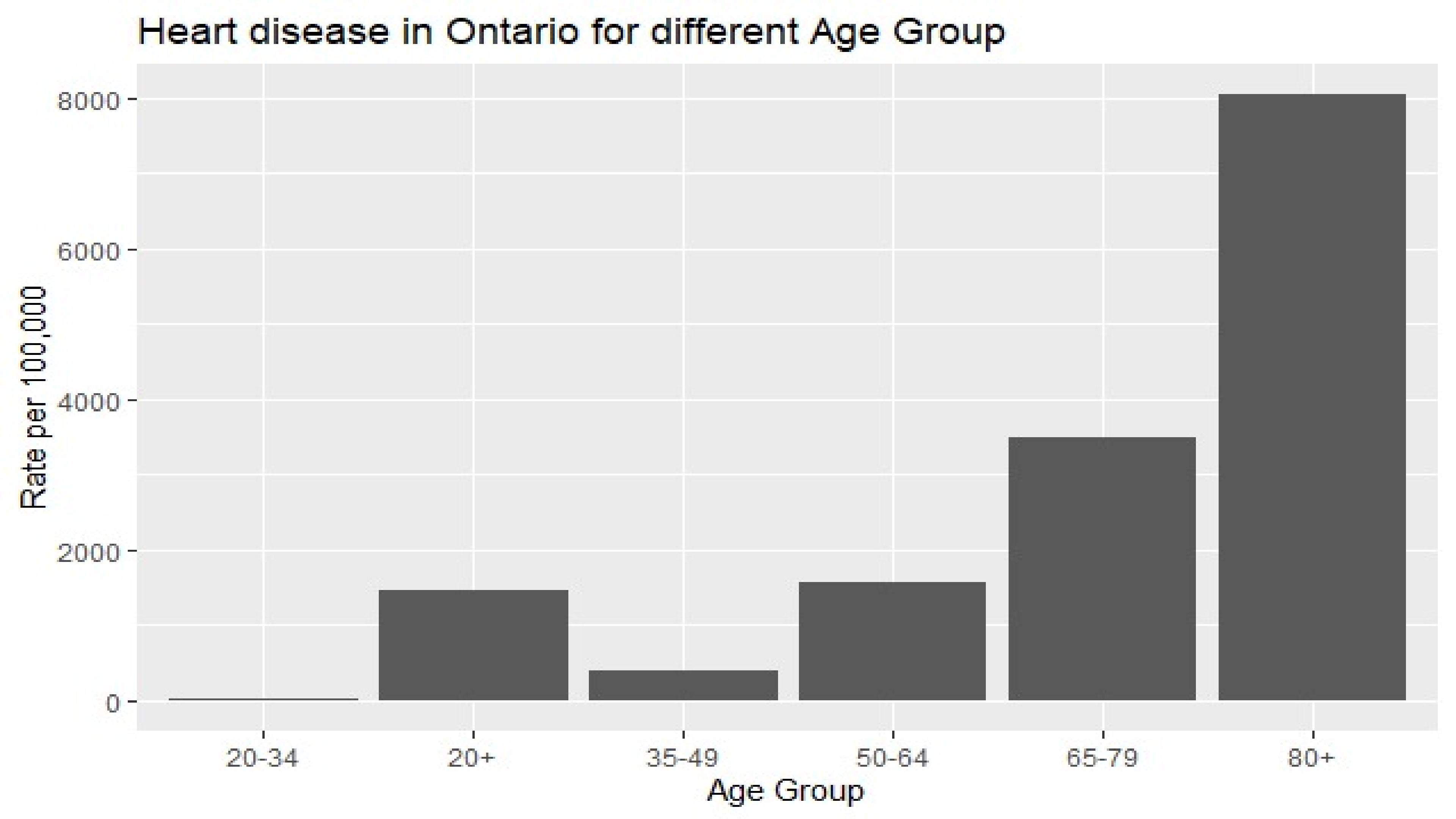
R 4.1.1 · ~/

```
+   labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
> View(total)
> View(total)
> total <- subset(heartdata, Sex="Both sexes", Age.Group=="20+", select = c("Age.Group","Sex","Year","Rat
e"))
> ggplot(total, aes(x=Year, y=Rate)) +
+   geom_bar(stat='identity') +
+   labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
> View(total)
> View(total)
> total <- subset(heartdata, Sex=="Both sexes", Age.Group=="20+", select = c("Age.Group","Sex","Year","Ra
te"))
Error in `[.data.frame`(x, r, vars, drop = drop) :
  object 'Age.Group' not found
> ggplot(bothSex, aes(x=Age.Group, y=Rate)) +
+   geom_bar(stat='identity') +
+   labs(title = "Heart disease in Ontario for both gender",x="Age Group",y="Rate per 100,000")
> total <- subset(heartdata, Sex=="Both sexes", select = c("Age.Group","Sex","Year","Rate"))
>
> totalYear = filter(total, Age.Group =="20+" )
>
> ggplot(total, aes(x=Year, y=Rate)) +
+   geom_bar(stat='identity') +
+   labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
> totalYear <- filter(total, Age.Group =="20+" )
> View(totalYear)
> View(totalYear)
> ggplot(totalYear, aes(x=Year, y=Rate)) +
+   geom_bar(stat='identity') +
+   labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
>
```


Yearly total number of heart disease in Ontario
Rate per 100,000
0
50
100
150
200
2011
2013
2015
Year

Heart Diseases in Ontario by Age Group.





Heart disease in Ontario for different Age Group
8000
6000
4000
2000
0
Rate per 100,000
20-34
20+
35-49
50-64
65-79
80+
Age Group

Conclusion

- From the Histogram chart, we see that the % of heart disease is slightly higher in case of male than that of female. The rate of heart disease decreased from 2011 to 2015 gradually.

- It is interesting to find that middle age group has lower rate of heart disease as compared to aged groups (80+).
- It is also noteworthy over younger ages groups (20+) have higher tendency of heart disease possibly due to food habits.

- Overall, OHMA is trying to spread awareness in people about heart disease and in our analysis. People are now getting more conscious about their diet and exercise routine to live healthy life

Thanks for attending.

Questions ?